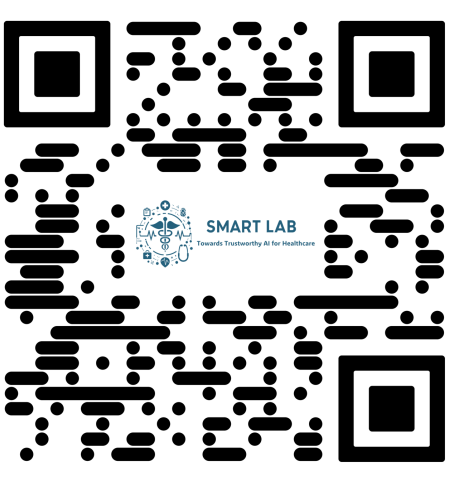


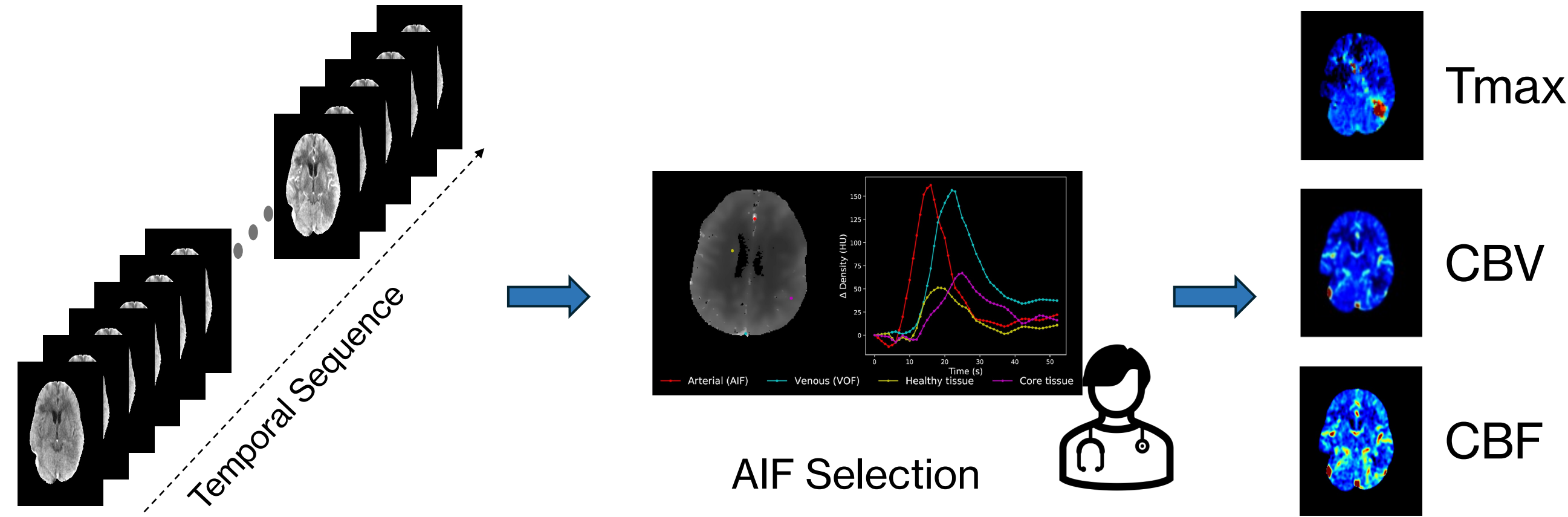


Progressive Knowledge Distillation for Automatic Perfusion Parameter Maps Generation from Low Temporal Resolution CT Perfusion Images

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Background: Computed Tomography Perfusion (CTP)



Full Temporal Resolution
CTP Scan

Perfusion Parameter
Maps

- Perfusion Parameter Maps (PPMs) derived from **Computed Tomography Perfusion (CTP)** images, play a critical role diagnosing cerebral pathologies including **stroke, tumors, and vascular disorders**.
- While **Perfusion Parameter Maps (PPMs)** hold significant clinical value, the current process of obtaining such images pose **several challenges**
 - High radiation exposure** during the CTP scanning
 - Dependency of physician-provided** Arterial Input Function (AIF) information.

Methodology

- Multitask Learning** is leveraged to take advantage of the intercorrelated information between PPMs.
- Knowledge Distillation** begins with intermediate resolutions **progresses to** high resolution teacher network to reduce the large initial knowledge gap.
- When the loss is stale for certain patience level the teacher network will be switched to the next teacher network. This ensures **stable learning and Knowledge Distillation** process.

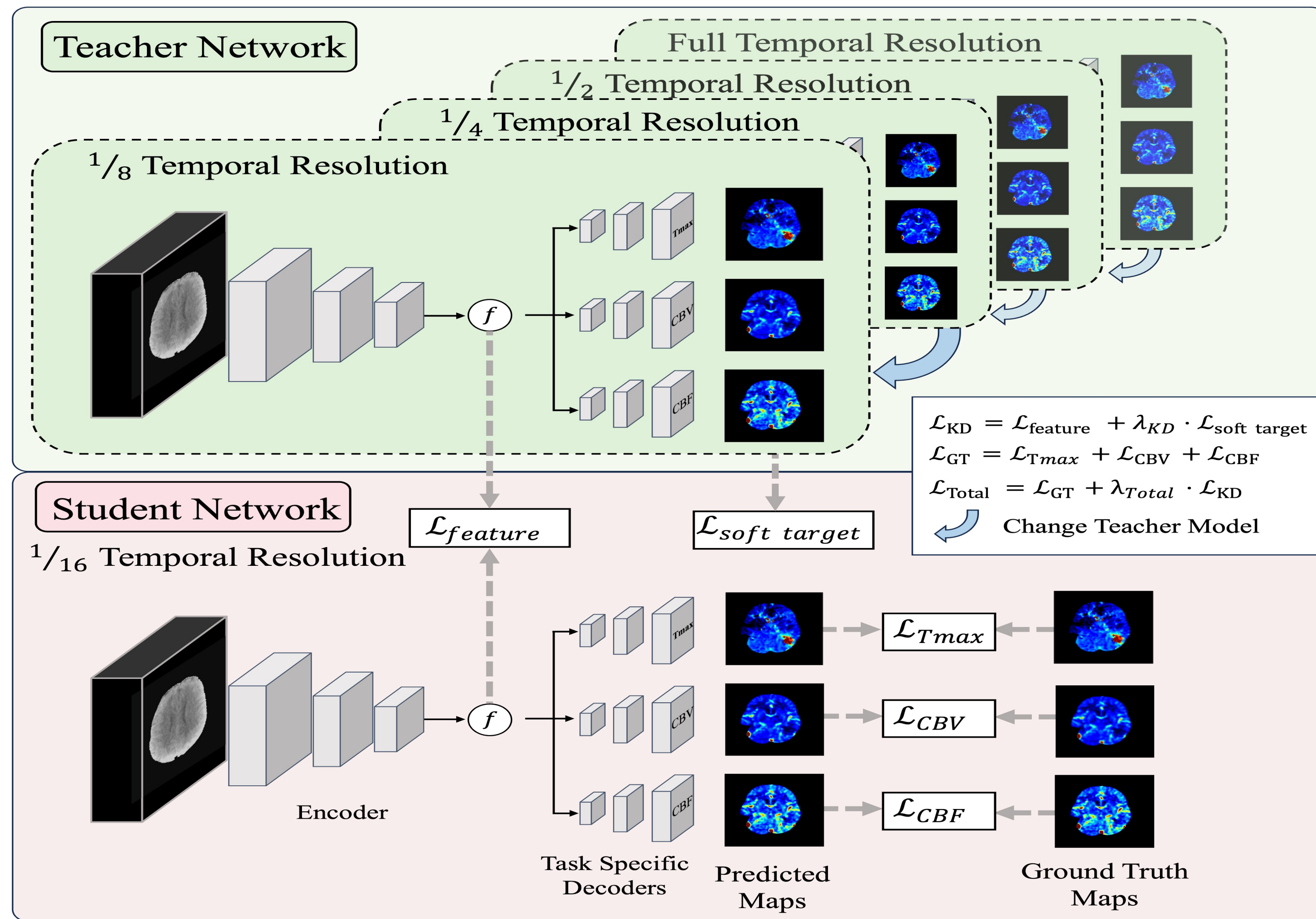


Fig. 1. Progressive Knowledge Distillation framework consists of two main parts: the teacher networks and the student network of the same 3D V-Net architecture (1 encoder and 3 decoders)

Contribution

1. Automate PPMs Generation with Multi-task Learning

- Traditional PPM generation relies on **manual Arterial Input Function (AIF) selection**.
- Our approach **automates this process without AIF, leveraging Multi-task Learning** to capitalize on the inherent intercorrelation of PPMs derived from cerebral blood flow dynamics.

2. Reduce Radiation Exposure in CT Perfusion (CTP) with Knowledge Distillation

- CTP scans expose patients to high radiation levels due to **long range of scanning**.
- Previous studies have attempted to reduce radiation exposure by **increasing the scanning interval resulting in a low temporal resolution** CTP scan.
- We employ **Knowledge Distillation** to bridge the gap between full and low temporal resolution CTP scans, maintaining accuracy while reducing radiation exposure.

Results

	Temporal Resolution	Tmax			CBV			CBF		
		RMSE ↓	SSIM ↑	PSNR ↑	RMSE ↓	SSIM ↑	PSNR ↑	RMSE ↓	SSIM ↑	PSNR ↑
Ours	Full	1.201	0.913	27.965	8.892	0.918	28.167	48.163	0.932	27.693
UNET-GAN-Assisted [1]	Full	1.355	0.903	26.646	9.144	0.917	27.893	49.680	0.929	27.413
UniToBrain [2]	Full	1.314	0.900	26.834	9.117	0.916	27.798	50.196	0.926	27.051
Ours w/ PKD	1/8	1.033	0.923	29.709	9.000	0.915	28.015	48.207	0.928	27.729
Ours w/o PKD	1/8	1.516	0.869	25.023	9.536	0.912	27.467	52.536	0.921	26.909
UNET-GAN-Assisted [1]	1/8	1.769	0.852	23.662	9.662	0.908	27.252	58.508	0.909	25.676
UniToBrain [2]	1/8	1.460	0.885	25.728	9.778	0.895	27.040	51.565	0.922	26.829
Ours w/ PKD	1/16	1.436	0.879	25.790	9.004	0.916	28.061	49.639	0.926	27.430
Ours w/o PKD	1/16	1.516	0.869	25.023	9.536	0.912	27.467	52.536	0.921	26.909
Ours w/ DGKD [5]	1/16	1.721	0.859	24.279	10.661	0.893	26.806	62.742	0.900	25.593
Ours w/ KD [3]	1/16	1.511	0.870	25.463	9.594	0.910	27.506	52.298	0.922	26.820
Ours w/ MTKD [6]	1/16	1.464	0.875	25.661	9.518	0.910	27.403	51.502	0.922	26.897
Ours w/ TAKD [4]	1/16	1.492	0.871	25.356	9.273	0.913	27.842	52.217	0.921	27.095
UNET-GAN-Assisted [1]	1/16	2.171	0.842	21.782	10.049	0.898	26.889	57.849	0.907	25.691
UniToBrain [2]	1/16	1.726	0.838	24.044	10.321	0.893	26.483	59.901	0.884	25.310

Table 1. Results of perfusion parameter map estimation. **Bolded** Values indicate the best performance. Underlined values indicate the best performance across all rows.

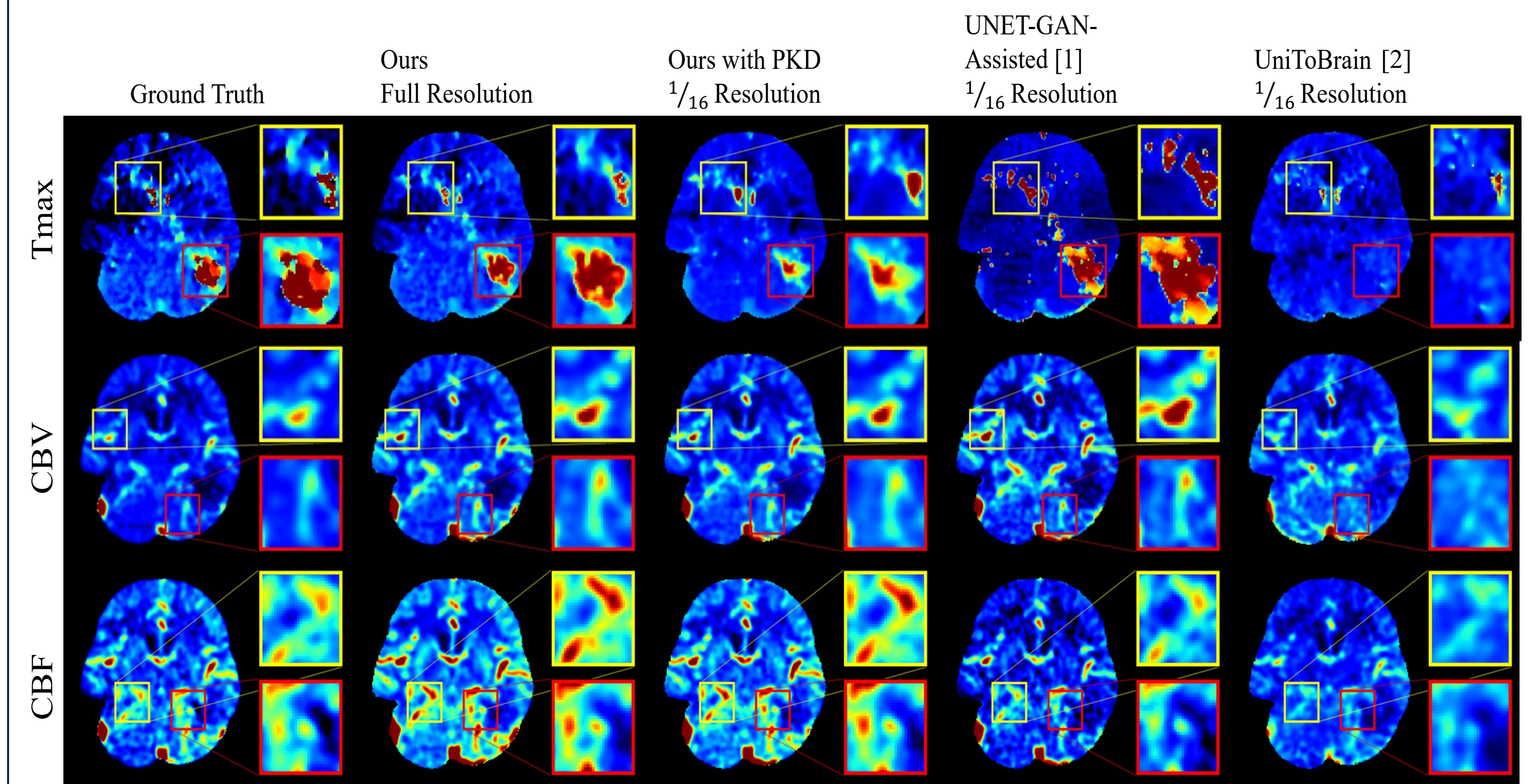


Fig. 2. Visual comparison of the generated perfusion parameter maps. **Red** and **Yellow** boxed areas represent lesions with the first and second largest disagreement between maps

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